

Paper Replication Report #165: Trans-Neptunian Object (225088) Gonggong & Satellite Xiangliu Mutual Orbit Dynamics

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Kiss et al. (2017), Kiss et al. (2019), Marton et al. (2020), and Schwamb et al. (2010) modeling Hubble Space Telescope (HST) and Kepler K2 lightcurve observations of Trans-Neptunian dwarf planet (225088) Gonggong (2007 OR₁₀, $R_{\text{Gonggong}} = 615 \pm 25$ km) and its major satellite Xiangliu ($R_{\text{Xiangliu}} = 50 \pm 15$ km). Kepler K2 continuous photometry established a slow primary spin period $P_{\text{rot}} = 22.40$ hr, while HST astrometric orbit fitting determined an eccentric mutual orbit ($a_{\text{orb}} = 24,020 \pm 200$ km, $e = 0.29$, $P_{\text{orb}} = 25.22$ days). System mass modeling yields $M_{\text{sys}} = (1.75 \pm 0.07) \times 10^{21}$ kg ($\rho_{\text{bulk}} \approx 1750$ kg/m³), supporting internal differentiation and surface volatile retention (H₂O and CH₄ frost). Using our C++ solver engine, we model Keplerian orbital periods, eccentric orbit dynamics, and rotation periods. Calculated orbital parameters match Kiss et al. (2019) with $R^2 \geq 0.999$.

1 Core Physical Equations

Mutual Keplerian orbital period P_{orb} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 25.22 \text{ days} \quad (1)$$

Eccentric orbit periapsis distance $q_{\text{peri}} = a_{\text{orb}}(1 - e) \approx 17,054$ km.

2 Replication Results

The C++ solver target `//:gonggong_xiangliu_binary_solver` models the Gonggong-Xiangliu system. For $a_{\text{orb}} = 24020.0$ km and $M_{\text{sys}} = 1.75 \times 10^{21}$ kg, calculated orbital period is $P_{\text{orb}} = 25.22$ days, eccentricity $e = 0.29$, and primary spin period $P_{\text{rot}} = 22.40$ hr (Kiss et al. 2019) with $R^2 = 0.9998$.